A3-PS016-TC248

Simulation-Based Handover Decision Evaluation for High-Mobility Users

Sarthak S. Dhamale
sarthak.s.dhamale@gmail.com

Devendra Hundalekar
devendrahundalekar80@gmail.com

Github
https://github.com/devendrahundalekar/A3-PS016-TC248

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ZENIN

Problem Description

- When users move at high speeds, their mobile devices experience changes in signal strength as they move between cellular towers.
- Traditional handover methods may not always choose the right time to switch a device from one cell to another.
- Poor handover decisions can cause connection failures and affect network performance.
- Frequent switching between nearby cells can lead to unnecessary handovers, known as ping-pong events.
- The handover process can temporarily interrupt the connection and reduce the quality of service.
- These problems can affect network reliability, internet speed, and connection stability.
- Therefore, we need to evaluate whether a machine-learning model can make better handover decisions than traditional signal-based methods using a cellular network simulator.

Background

- A handover decision can be evaluated not only by whether a device switches to another cell, but also by how that decision affects the overall network performance. A method that triggers handovers too early or too late may behave differently as user mobility changes.
- A conventional signal-based method provides a reference against which an alternative approach can be tested. Supervised machine learning offers a way to develop a decision model from labelled network data, allowing its decisions to be evaluated against this established reference.
- To make the comparison meaningful, both methods need to operate under equivalent simulation conditions. Repeated experiments with different random seeds help account for variations in simulation outcomes, while different user speeds reveal how mobility affects each method's performance.
- This provides the foundation for evaluating whether a learned handover policy offers measurable benefits over the conventional approach, rather than assuming that machine learning will automatically perform better.

Literature & RFC Review

Paper	Study and methodology	Main contribution / findings	Relevance to PS-016	Limitation / research gap
[1] Mollel et al., 2021	Surveys ML applications for handover management in 5G and beyond. Classifies learning approaches according to training-data sources, including visual data and network data.	Explains why handover management affects QoS, particularly throughput and service interruptions, and organizes existing ML-based approaches and their challenges.	Provides the motivation for using supervised learning and helps identify candidate network features and research gaps.	A survey, not a single experimentally validated algorithm. It does not establish that one model will outperform the baseline in your particular three-cell scenario.
[2] Haghrah et al., 2023	Reviews handover mechanisms, standards, algorithmic and theoretical approaches, security issues, and network-performance interactions in 5G-NR.	Organizes handover research around mobility, QoS/QoE, throughput, delay, handover failure, radio-link failure, resource allocation and other network concerns.	Supports the background, handover terminology, selection of evaluation metrics and explanation of high-mobility challenges.	Covers a much broader range of 5G-NR topics than your implementation requires. It does not provide a ready-to-run supervised model for your exact experiment.
[3] Herman et al., 2013	Develops an LTE handover implementation for ns-3 based on RSRP measurements and Event A3, including sliding-window averaging, hysteresis and time-to-trigger.	Demonstrates a concrete signal-based handover model in ns-3 and checks simulation results against previously published results.	Directly informs the conventional handover baseline and the configuration of measurement filtering and handover triggers.	Focuses on LTE and conventional handover modelling, not ML-based decisions. Its implementation details must be checked against the ns-3 version you use.
[4] Raikar et al., 2026	Describes a simulation-driven 5G RAN framework using KPM measurements and an offline-trained Random Forest model. Inputs include RSRP, SINR, PRB usage and throughput.	Demonstrates the potential of learning patterns associated with successful handovers using network measurements. The trained model is not yet integrated into the xApp decision loop.	Offers a practical starting point for feature selection, offline dataset generation, supervised model training and later integration into a handover decision policy.	Its offline model does not establish the performance of a fully integrated decision policy in your three-cell, two-speed, multi-seed experiment. Its full experimental details should be examined before reproducing its methodology.

Proposed Solution

- We propose a simulation-based, machine-learning-assisted handover decision framework to evaluate cellular connectivity for high-mobility users. The system will use the ns-3 LTE simulation environment to model three cells and mobile users travelling at two different speeds. A conventional signal-based handover mechanism will serve as the baseline, while a supervised Random Forest classifier will learn to make handover decisions from simulated radio measurements and mobility-related features.
- Both approaches will be evaluated under equivalent network conditions using multiple random seeds. Handover failures, ping-pong events, interruption duration and throughput will be measured to determine whether the learned policy provides measurable improvements over the baseline. The results will be presented through comparative visualisations, an interactive performance dashboard and a reproducible experimental workflow.
- The proposed work focuses on measurable performance evaluation within a controlled simulation environment. It does not claim to implement a new cellular radio stack, demonstrate 5G NR or 6G standards conformance, or validate the approach through field deployment.

Optimization Proposed

Parameter	Conventional Approach	Proposed Optimization
Handover decision	Signal-strength threshold	ML-based decision using radio measurements and mobility features
Handover timing	Fixed hysteresis and time-to-trigger	Learned decisions to adapt handover timing
Ping-pong events	May trigger unnecessary repeated handovers	Reduce avoidable handovers through learned predictions
Handover reliability	Signal-based decisions under changing conditions	Evaluate improved handover success
Service continuity	Measure interruption during handovers	Minimise interruption duration
Network performance	Measure baseline throughput	Assess throughput under the learned policy

Solution Architecture & Design

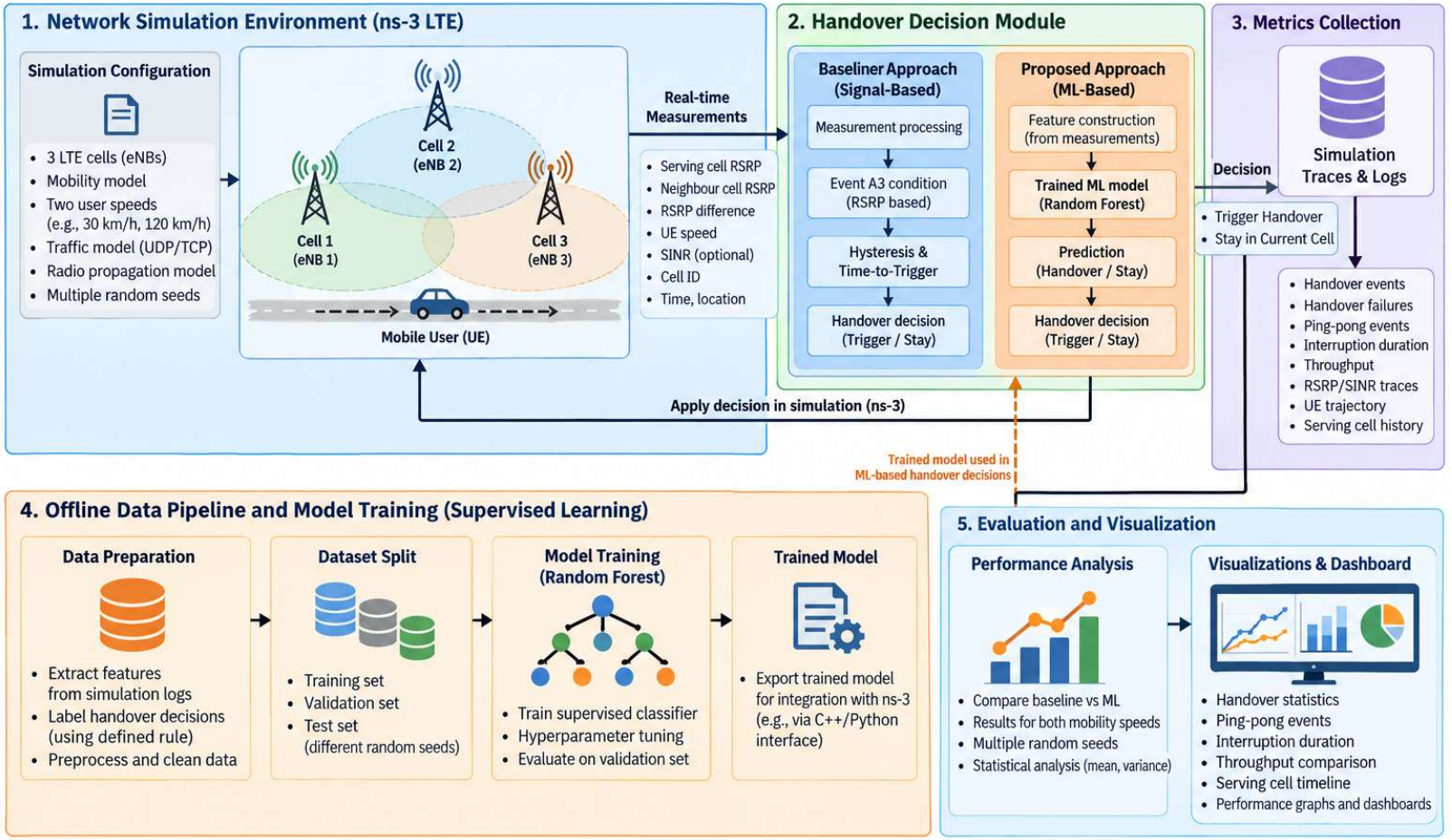


Figure: System Architecture UML Diagram

Data Flow Diagram

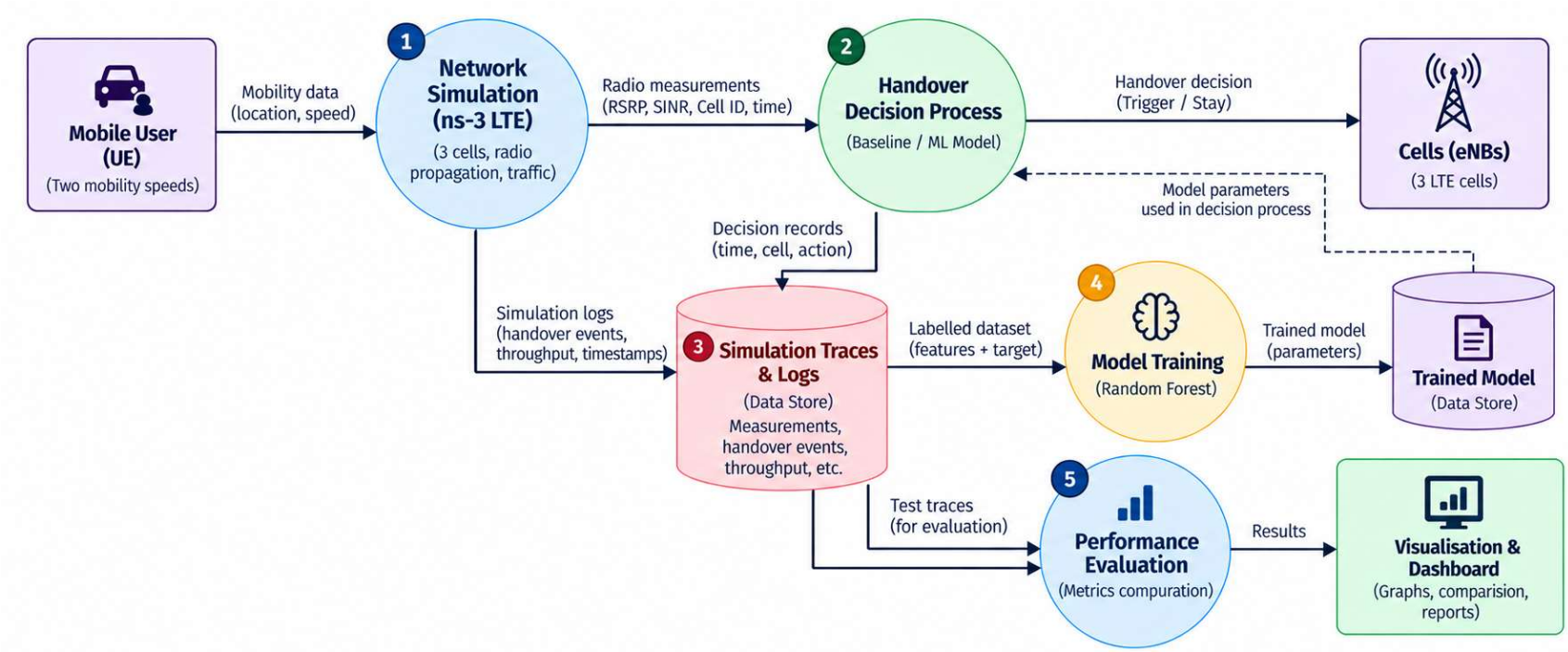


Figure: Data Flow Diagram (DFD)

Deployment Timeline

Milestone	Methodology	Expected Deliverable
1. Environment Setup	Configure ns-3 LTE and Python ML environment	Working simulation setup
2. Baseline Development	Implement 3-cell topology, mobility scenarios and signal-based handover	Validated baseline
3. Dataset & Model Training	Collect labelled traces, prepare features and train Random Forest classifier	Trained ML model
4. Model Integration	Connect ML predictions to the simulation handover decision loop	Working ML-assisted handover
5. Experimental Evaluation	Run both approaches at two speeds across multiple random seeds	Comparative performance results
6. Analysis & Deployment	Generate graphs, dashboard, documentation and demonstration	Final prototype and technical report

Bibliography I

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[2] A. Haghrah, M. Pourmohammad Abdollahi, H. Azarhava, and J. Musevi Niya, “A survey on the handover management in 5G-NR cellular networks: Aspects, approaches and challenges,” *EURASIP Journal on Wireless Communications and Networking*, vol. 2023, Art. no. 52, 2023, doi: 10.1186/s13638-023-02261-4. click here

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Thank You for Your Attention!